Video Quality Assessment

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A Ph.D. Dissertation Proposal

April 10, 2002

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Overview

Introduction

The digital video systems have many advantages over the existing analog methods. Digital video processing and transmission techniques offer an efficient and economical way of implementing a video communication system. Flexibility is probably one of the biggest advantages of this type of system.

Traditional analog quality measures have relied on the "constancy" of a video system's performance for different input scenes. Thus, one could inject a test pattern or test signal (e.g., a static multi-burst), measure some resulting system attribute (e.g., frequency response), and be relatively confident that the system would respond similarly for other video material (e.g., video with motion). A great deal of research has been performed to relate the traditional analog video quality measures (e.g., differential gain, differential phase, short time waveform distortion, etc.) to perceived changes in video quality. While the recent advent of video compression, storage, and transmission systems has not invalidated these traditional parameters, it has certainly made their connection with perceived video quality much more tenuous. Digital video systems adapt and change their behavior depending upon the input scene. Therefore, attempts to use input scenes that are different from what is actually used in-service can result in erroneous and misleading results.

Digital video systems introduce fundamentally different types of impairments than those created by analog reproduction methods. Examples of compression-related impairments are edge busyness, error blocks, localized smearing, jerkiness (i.e., jerky motion), and blocking/tiling. For digital video systems, the spatial and temporal information content of the source plays a crucial role in determining the amount of compression that is possible and hence the severity of the compression artifacts. The digital transmission channel's operating characteristics may also be dynamic (e.g., bit rate, dropped packets) and this will cause the received video quality to vary even when the information content of the input video is fixed.

Most applications of digital video are destined for human consumption. Human perception comes into play when trying to establish the degree to which a video can be compressed [1], figuring out how much data can be hidden in an image [2], deciding whether visual enhancements have provided an actual benefit [3], and choosing a set of image features to use for segmentation and indexing [4]. Over the years, measurement of the performance of various algorithms has been largely limited to a few objective measures such as the mean absolute error (MAE), mean square error (MSE), and peak signal-to-noise (PSNR). Although this approach is fairly standard in published literature, it suffers from one major weakness. The outputs of these measures do not always correspond well with human judgments of quality [1]. In particular, it is difficult to compare the outputs of these measures for different images and different transformations.

At present, no physical measure can accurately predict the visibility and annoyance for compression artifacts. The only way to determine visibility and annoyance is to measure them directly in psychophysical experiments with human subjects. However, psychophysical experiments are too expensive and time-consuming to be a practical method for measuring image quality. Therefore, there is a great need for an objective quality metric, which correlates well with the human perception of quality. This metric should be, at the same time, fast and reliable.

Background on Quality

Although quality is a rather difficult concept to grasp, two simple concepts about quality are widely accepted. First, quality always implies a comparison between two or more objects. The comparison may be direct (A is better than B) or indirect (A is good). Second, quality must be measured using an open-ended scale. There is always a chance of something being better or worse.

These two simple concepts have a big impact on how video quality is measured. When a reference is not available, individuals will judge the quality of a video based on an internal reference. To remove the variation between internal references of different persons, explicit reference videos are often shown. To make the scales open-ended, both the subjective and objective quality scales should allow for responses to test videos that are both better (increasing quality) and worse (decreasing quality) than the reference video [6].

Frequently subjective tests measure impairment scores instead of quality scores to avoid double-ended scales. Impairment is a subjective measure of the degradation of a video. The reference corresponds to the best possible quality in an impairment measurement experiment and any difference from this reference is considered to be a decrease in quality or an increase in impairment. Observers are generally asked to rate video quality in terms of annoyance, where annoyance is a measure of how "bad" the observer think the impairment is. The annoyance value is assumed to increase with the strength of the impairment.

Objective measures of impairment are called fidelity measures [7] and are based on the difference between reference and test videos. By definition, fidelity measures require information about the reference video, which can be very restrictive. The fidelity measures are categorized according to the amount of reference information they require. Full-reference fidelity measures require the entire reference video and may not be very practical for real time transmission applications. Reduced-reference fidelity uses a reduced size version of the reference, while non-reference fidelity measures use no reference at all.

Two frequently used fidelity measures are the MSE and the PSNR. These measures are based on the sum of all differences between the video and the reference. No distinction is made between impairments that humans can and cannot see. Although simple and fairly standard in the literature, these measures do not always correlate well with human judgments of quality. Attempts have been made to try develop perceived fidelity measures that correspond to the perceived differences between an original and a transformed video. These measures are based on models of the human visual system [1, 8, 9, 10].

A defect is defined as being a perceived flaw introduced into an image or video during capture, transmission, storage, and/or display as well as by any image processing algorithm (e.g. enhancement, compression) that may be applied along the way. Defects may be decomposed into defect features. For example, subjects may report defects with varying strengths of blockiness, loss-of-detail, ringing, and mosquito noise features after MPEG-2 compression [11]. Although most defects have more than one feature, it is possible to produce defects that are relatively pure.

Error component measures are objective measures that attempt to predict the perceived defect feature strengths. A good error component measure would be able to predict the perceived strength of a single defect feature. A good overall measure would predict perceived impairment by combining error component outputs [12].

Problem Statement

The main objective of this work is to develop objective quality metrics that are closely related to quality as experienced by human viewers. Ideally, this metric would correlate well with subjective responses independent of the video content and type of the corruption introduced. The algorithm used to calculate such a metric should be fast, efficient and need a minimal knowledge of the original (reference) video.

A good way to acquire the knowledge necessary to develop such metric is by measuring and analyzing the responses of human observers to defects in video sequences. With this objective, we intend to perform a series of subjective experiments and develop several specific-purpose objective measures. The data gathered during the experiment is used to check assumptions and develop models.

This work is divided in two parts. In the first part, we develop algorithms for measuring the detectability and annoyance of the most important types artifacts found in digital video after transmission/storage, compression, enhancement or some other type of processing. To tackle this problem, we intend to isolate, as much as possible, the different types of artifacts and study their main characteristics. An ideal end-result of this part of the work would be a series of objective measures that can predict the annoyance of a variety of impairments and an overall algorithm that combines the individual measures into an overall quality score.

As part of this effort, we intend to develop methods for synthetically simulating various artifacts. Good synthetic artifacts can be used to simplify psychophysical test design and guide the development of objective measures of impairment strength. Psychophysical experiments will be performed to validate and carefully study the different types of artifacts. Details of this first part of the work will be described in Section 2 of this proposal.

The second part of the work involves the design of a non-reference objective metric for estimating impairments introduced during capture, transmission, storage, and display as well as by some transformation like compression or enhancement. This metric can be used in applications requiring video transmission in which cases, the full and reduced reference algorithms are



(a) (b) (c)

Figure 1: The three regions selected from this video are shown in reverse video. They were (a) the sky, (b) the houses, and (c) the garden. In each test sequence, only one region was corrupted.

not practical because of the high cost of sending any side information about the reference video. Our approach makes use of data hiding techniques to transmit the necessary reference information embedded in the video signal. Our preliminary work has shown that a digital mark can act as a reference for an impairment measurement. This system needs to be further developed and tested with psychophysical data. Details of this second part of the work is described in Section 3 of this proposal.

Experimental Methods

The Image Processing Laboratory (IPLAB) at UCSB, in conjunction with the Psychology Department, has been performing experiments on video quality for the last three years. The majority of the experiments have been aimed at measuring detection thresholds and perceived impairment, in terms of annoyance, for digital video systems impairments. Our test subjects were drawn from a pool of students in the introductory psychology class at UCSB. The students are thought to be relatively naive concerning video artifacts and the associated terminology.

Basic Methodology

The normal approach to subjective quality testing is to degrade a video by a variable amount and ask the test subjects for a quality/impairment rating [6, 14]. The degradation is usually applied to the entire video. In this research we have been using an experiment paradigm that measures the annoyance value of brief, spatially limited artifacts in video [5]. We degrade one specific region of the video for a short time interval. The rest of the video clip is left in its original state. Figure 1 shows an example of the regions degraded for the video "flower-garden" in one of our experiments.

The test sequences used in the experiments are generated from a set of original video sequences of assumed high quality. The videos are all five seconds long and contain scenes typical of normal television. The number of original sequences should be maximized because the appearance of the artifacts, and therefore their visibility and annoyance depend on the content of the video. Varying the content also helps to keep the subjects interested. For each defect zone, a corresponding time interval is chosen. The artifact duration is limited for the same reasons that the artifact size is limited. The first and fifth seconds of each video are never altered to avoid end effects. The regions and time intervals where defects appear are varied to prevent the test subjects from learning the locations.

To create the test sequences we first generate sequences with high level of impairment. The artifacts can be real or synthetic. Then, to generate the test sequences we combine the original video and the impaired video in different proportions. By varying their relative weights, we could weaken the artifact (allowing the original to dominate), strengthen the artifact (allowing the artifact to dominate), or even exaggerate the artifact (boost the difference between the artifact and original). The basic formula is:

$$Y = X_0 \cdot (1 - r) + r \cdot X_1, \qquad (1)$$

where Y is the result, X_0 is the original, X_1 is the sequence with the combined artifacts, and r is the scaling factor ($0 \le r \le 1$).

The artifacts are added only to the defect zones. All other areas are kept untouched. This procedure guarantees that the appearance of the artifacts we are testing do not change over the test sequences, only their strength varies.

Apparatus

For our experiments, the test sequences are stored on the hard disk of an NEC server. Each sequence requires just about 100 megabytes of storage and the entire test set for one experiment uses about 10 gigabytes of disk space. The server uses a single 18-gigabyte SCSI hard drive to store the test sequences. With this configuration, each test sequence can be loaded and displayed in six to eight seconds.

Each video is displayed using a subset of the PC cards normally provided with the Tektronix PQA-200 picture quality analyzer. A generator card is used to locally store the video and stream it out in a serial digital (SDI) component format. The test sequence length is limited to five seconds by the generator card. The generator card output is sent to a decoder card that converts the SDI signal into an analog NTSC S-video output. The analog output is then displayed on a Sony PVM-1343 monitor. The result is five seconds worth of broadcast quality, full-resolution, NTSC video.

In addition to storing the video sequences, the server is also used to run the experiment and collect data. A special-purpose program records each subject's name, displays the video clips, and runs the experiment. After each test sequence is shown, the experiment program displays a series of questions on a computer monitor and records the subject's responses in a subject-specific data file. The questions can vary from experiment to experiment.

The experiments are run with one test subject at a time. The subjects are asked to wear any vision correction devices (glasses or contacts) that they would normally wear to watch television. Each subject is seated in front of the computer keyboard at one end of a table. Directly ahead of the subject is the Sony video monitor, located at or slightly below eye height for most subjects. The subjects are initially positioned at a distance of four screen heights (80 cm) from the video monitor. The subjects are asked not to move too much during the experiment, although they are not prevented from doing so. The computer monitor is located to the right of the keyboard and closer to the test subject.

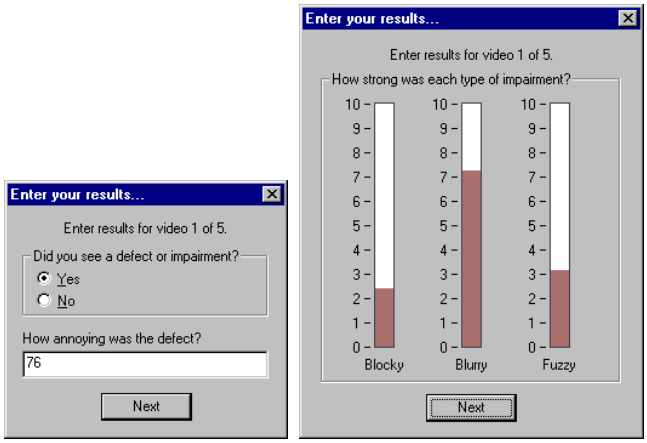
The test sequences appear on the video monitor directly in front of the viewer. After each sequence, questions appear on the computer monitor to the right. This setup requires the subject to switch attention from the video monitor to the computer display. However, because the same questions are always asked after each sequence, the subjects can quickly memorize the questions and rarely need to look at the computer monitor.

Procedure

The course of each experimental session goes through four stages: instructions, training, main experiment, and interview. In the first stage, the subject is verbally given instructions. In the second stage, training sequences are shown to the subject. The training sequences represent the impairment extremes for a given experiment and are used to establish the annoyance value range. The training trials are identical to the experimental trials, except that no data is recorded. The training is also used to familiarize the subject with the experiment. Eight to fifteen trials are included in the session, depending on the complexity of the experiment, to allow the subjects' responses to stabilize before the main experiment begins.

After the training, the main experiment is performed with the complete set of test sequences. In the main experiment, subjects enter data using the dialog box like the one shown in Figure 2(a). The first question on the dialog ox is always whether an impairment/ defect was seen. If the answer is 'no', the subject can hit next and the following test sequence is played. If the answer is 'yes', the subject is asked to enter a value, between o and a 100, proportional to the annoyance caused by the defect or impairment. The subjects are asked to set a value of 100 for the worst defect they see in the training clips. During the main experiment, we ask them to give a 100 value to any video they think is as annoying as those worst videos in the training session. Any half as bad should be given 50, and any twice as bad 200 and etc. Although we try to include the worst test sequences in the sample set, we acknowledge the fact that the subjects might find some of the other tests clips to be worse and specifically instruct them to go above 100 in that case.

In addition to asking for detection decisions and annoyance values, some variations of this experiment may ask for further information about the defect, like for example its position or its appearance. Figure 2(b) and (c) depicts examples of dialog boxes used in past experiments to ask further questions to the subject.



(a) (b)

Figure 2. Dialog boxes used for subjective data entry in (a) detection threshold and (b) defect feature analysis experiments.

At the end experiment, we ask the test subjects for qualitative descriptions of the defects that were seen. The qualitative descriptions are useful for categorizing the defect features seen in each experiment and help in the design of defect feature analysis experiments.

Data Analysis

The data collected during the experiment, in the form of observers' score for each test sequence, must be condensed by statistical techniques to yield results in graphical and/or numerical/formulae/algorithms. The experiments were designed specifically to measure detection thresholds as well as the annoyance values for each test condition. In our analysis, one test condition generally refers to a set of test sequences with the same defect zone, created from the same original, but with varying impairment scaling factors (r in Eq. (1)).

To perform our data analysis, two measures for each test condition are needed: the total squared error (TSE) and the Mean Observer's Score (MOS). The TSE is our initial objective error measure and is defined as:

$$\mathrm{TSE} = \sum (M - X)^2 , \tag{2}$$

where M is the test sequence and X is the original sequence. The MOS is our subjective error measure and is calculated by averaging the annoyance level for all observers:

$$\mathrm{MOS_{j,k}} = {}^{1}\!/_{N} \sum_{i=1}^{N} A(i, j, k), \tag{3}$$

where $A\ (i,j,k)$ is the annoyance level reported by the i-th observer for the weight j of the impairment and test condition k. N is the total number of observers. In most of our experiments we have used six different scaling factors for each test condition.

The threshold is defined as the log error energy such that the artifact was seen by 50% of our subjects. In our experiments, we estimate the probability of detecting each artifact by counting the number of people who detected the artifact and dividing by the number of observations. The probability as a function of the $\log_{10}$ TSE (Eq. (2)) is fitted using the Weibull function, which has an S-shape similar to our data and is defined as:

$$P(x) = 1 - 2^{-(S \cdot x)^k} , \tag{4}$$

where $P(x)$ is the probability of detection, x is $\log_{10}$TSE, $1/S$ is the 50% detection threshold in log total squared error, and k is a constant that indicates the steepness of the function.

We use the standard methods [6] for analyzing the annoyance judgments provided by the test subjects and computed the mean annoyance values. The mean annoyance values for each test sequence were fitted with the standard logistic function [6]:

$$y = y_{\min} + \frac{(y_{\max} - y_{\min})}{\left(1 + \exp\left(-{}^{(x - \bar{x})}\!/_{|\beta|}\right)\right)}, \tag{5}$$

where y is the predicted annoyance and x is the log10 (TSE). The parameters $y_{\min}$ and $y_{\max}$ establish the limits of the annoyance value range. The parameter $\bar{x}$ translates the curve in the x-direction and the parameter β is inversely related to the steepness of the curve.

After Eqs. (4) and (5) are estimated using curve fitting procedures, we plot the corresponding psychometric function and annoyance curves for each of the test conditions. We also plot the overall curves fitted for all the test conditions.

Part I: Objective Quality Metric based on Impairments Measurements

Digital video systems introduce different types of impairments, making it difficult to design an objective quality metric capable of measuring all of them. A feasible approach is to develop individual error component measures for each of the impairments introduced during capture, transmission, storage, and display as well as by some transformation like compression or enhancement [13]. In Table 1 a brief description of the most common impairments found in digital video systems is given. The error component measures can be later combined to generate an overall objective quality metric, according to the perceptual impact of each one of the impairments. Figure 3 shows the general block diagram of such system. The main goal of this part of the work is to design a practical and efficient way of implementing this system.

The advantage of this approach is that the algorithms destined for measuring each one of the impairments generally are not required to be full-reference algorithms. Measuring individual types of artifacts can be much faster and more accurate than trying to measure the overall objective metric all at once. The algorithm can be even more efficient by using only the most important impairments to the application in mind. Another advantage of this approach is the possibility of identifying specific problems of the digital system through the individual error component measures. The presence of certain characteristic impairments may indicate which part of the system is more sensitive.

Many picture quality models have been proposed, but little work has been done on studying and characterizing the individual impairments found in digital video applications. A study of the artifacts and their characteristics is therefore necessary since we do not yet have a good understanding of how the perceived quality of impaired digital video images depends on their physical properties. One approach for studying impairments is to work with synthetic artifacts that look like "real" impairments, yet are simpler, purer, and easier to describe. This approach is promising because of the degree of control it offers with respect to the amplitude, distribution, and mixture of different types of impairments. Such impairments are necessary components of the kind of reference impairment system recommended by the ITU-T for the measurement of image quality [11] and offer advantages for experimental research on video quality. They would make possible, for example, to study the importance of each type of impairment, according to the human visual system.

For the video quality measurement shown in Figure 3, intuition and ad hoc procedures can be used to obtain a set of features, which are meaningful and easily computed. These feature measurements can then be used to obtain the individual error measurements for each one of the impairments. The error measurements are then combined in order to obtain the overall objective

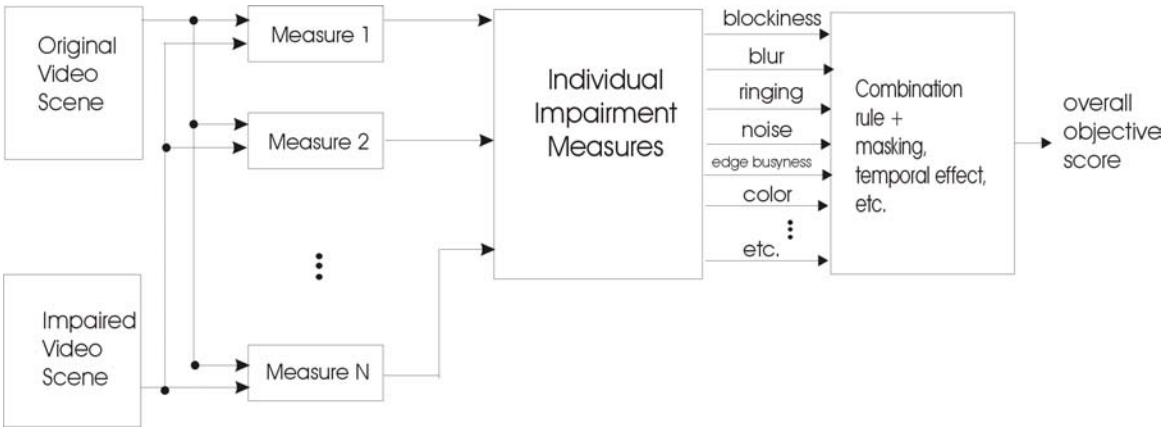


Figure 3: Block diagram for the reduced-reference objective metric based on error measurements.

Table 1. Summary of the most important impairments found in digital video systems [13].

Type of impairment	Description
Blockiness (Gibbs effect)	"Distortion of the image characterized by the appearance of an underlying block structure.
DCT basis and mosaic pattern effects	The DCT basis effect manifests itself as block bearing a distinct likeliness to one of the DCT basis image.
Blurriness	Global distortion over the entire image, characterized by reduced sharpness of edges and spatial detail.
Color bleeding	Blur in color components. It's more pronounced at the edges between areas with high color contrast and is visually perceived as a running or smearing of color.
False edges	Step-like graduation in intensity or color, in areas which had originally smooth transitions
Ringing	Appears as echoes of the hard edges in the picture or a rippling of an edge.
Edge busyness (ex. Mosquito noise)	The deterioration of motion video such that outlines of moving objects are showed with randomly activity.
Jerkiness	Motion that was originally smooth and continuous is perceived as a series of snapshots.
Ghosting	Delayed version of the image appear on the screen, either to the left or to the right.
Noise	An uncontrolled or unpredicted pattern of intensity fluctuations that is unwanted and does not contribute to the quality of a video image. Usually noise appears as either full or no signal from pixel concerned, which once interpreted by the device will appear as a spurious black, white or colored pixel.

quality score. This combination is generally not a simple addition. In fact, while combining these metrics we have to take account how important each of the impairments is, and also the effects of masking/facilitation that may exist between two or more artifacts. Besides that, while computing the overall quality objective metric we have also to take into account temporal aspects, attention spasms, eye movement, and many other problems. Experimenting with synthetic artifacts is extremely useful for solving this type of problem, because synthetic artifacts give us the possibility of carefully analyzing pure and mixed artifacts, their main characteristics and interactions.

Work done so far

We have implemented a method to generate two types of synthetic artifacts commonly found in compressed video sequences - blurriness and blockiness [15]. These two types of artifacts seem to be two of the most common visible artifacts found in compressed videos. The synthetic artifacts generated by the system satisfy the following criteria [11]:

- The appearance of these artifacts is similar to the impairments produced by compression algorithms.
- The impairment level range provides a wide range of video quality.
- The impairments can be combined in any proportion desired.
- The psychometric functions and the annoyance functions are similar to those for real artifacts.

Two psychophysical experiments were performed to measure the detection threshold and annoyance values of a set of synthetic artifacts. The Experiment I was performed on August 2001. Our goal was to test the viability of using synthetic artifacts for studying "real" MPEG-2 artifacts. For this first experiment, we chose to use a mix of blurry and blocky artifacts, which were as visually close as possible to the artifacts present in highly compressed MPEG-2 reconstructed videos. The data obtained with this experiment were then compared with the data obtained in an earlier experiment in which threshold and annoyance values of MPEG-2 artifacts were measured [5].

The Experiment II was performed on March 2002. In this last experiment we used pure and mixed blocky and blurry artifacts. Our goal was to measure their relative importance to the overall annoyance and their masking, facilitation properties. We are still in the process of analyzing the data form this experiment. Therefore, the results of this experiment will not be included in this proposal.

Experiment I

One of our goals for this experiment was to compare the visibility and annoyance value of synthetic artifacts with the real MPEG-2 artifacts. The MPEG-2 data used in this comparison was gathered in a previous experiment [5]. To validate our comparison, the same original sequences and defect zones used in the previous experiment were used in this experiment. The five original sequences used are: "Bus", "Cheerleader", "Flower-garden", "Football" and "Hockey". Artifacts were inserted into three defect zones in each at six different strengths. The total number of video sequences used in this experiment was 95, which includes 90 test sequences (5 originals times 3 defect zones times 6 scaling factors) plus the five original sequences. Each of these 95 sequences was presented once to each subject. The sequences were shown in a random order during the main experiment. The experimental followed the procedure described in the previous sections.

Generation of synthetic artifacts

The algorithm for producing the blocking artifacts is as follows: The first step is to calculate the average of each 8×8 block of the frame and of the 24×24-surrounding block, which has the current 8×8 block as a center. The next step is to calculate the difference, *D(i,j)*, between these two averages for each block. This difference is added to each block of the original frame *X(k,l)*:

$$A(k,l) = X(k,l) + D(i,j) \ , \qquad \begin{cases} i = round(k/8), & 1 \le i \le Rows/8 \\ j = round(l/8), & 1 \le j \le Cols/8 \end{cases} \tag{6}$$

where $i = round(k/8)$, with $1 \le i \le Rows/8$, and $j = round(l/8)$, with $1 \le i \le Cols/8$. *X* is the original frame, *A* is the frame with artifacts, *Rows* is the total number of rows of the frame, and *Cols* is the number of columns of the frame. The same algorithm was applied to luminance and the two color components of the video. Three other minor modifications of the artifacts were made: block average was matched to the original, borders of defect zones were faded, and saturation was controlled.

In our synthetic artifacts blurring was generated by applying a symmetric, two-dimensional FIR (Finite duration Impulse Response) low-pass filter through the digital image array. Different filters with varying cut-off frequencies can be used to allow control over the amount of blurriness introduced. We used a 5×5 mean filter in this experiment [16].

Table 2 Parameters of psychometric and annoyance functions for the synthetic and MPEG-2 artifacts inserted in the same regions of the sequences. Bold numbers indicate values that are statistically significant at $p = 0.05$.

	S		*k*		$\bar{x}$		β	
Sequences	Synthe.	MPEG	Synthe.	MPEG	Synthe.	MPEG	Synthe.	MPEG
BusBottom	0.26	0.27	19.15	9.60	4.39	4.08	0.27	0.45
BusMiddle	0.27	0.29	14.96	26.28	4.18	3.93	0.22	0.25
BusTop	0.28	0.32	10.18	12.71				
CheerBottom	0.28	0.27	19.71	13.03	4.18	4.11	0.28	0.33
CheerMiddle	0.30	0.28	13.12	18.30	3.92	4.02	0.27	0.45
CheerTop	0.28	0.27	10.57	19.27	4.42	4.39	0.34	0.46
FlowerGarden	0.24	0.25	17.09	14.98	4.62	4.59	0.29	0.37
FlowerHouses	0.29	0.28	12.18	15.71	3.99	4.03	0.26	0.29
FlowerSky	0.28	0.28	16.44	11.65				
FootballLeft					3.87	3.79	0.38	0.28
FootballMiddle	0.32	0.32	15.77	22.02	3.68	3.42	0.25	0.25
FootballRight					3.94	3.53	0.40	0.36
HockeyLeft	0.32	0.33	11.11	18.01	3.58	3.46	0.30	0.39
HockeyMiddle	0.33	0.36	19.79	17.78	3.55	3.37	0.25	0.30
HockeyRight					3.55	3.45	0.28	0.27
Mean	0.288	0.293	15.006	16.612	3.99	3.859	0.292	0.342
T-test, p =	0.281		0.414		**0.007**		**0.043**	
Correlation	**0.831**		-0.252		**0.930**		0.262	

The artifacts that we used were a mixture of blurry and blocky artifacts, which were visually similar to the artifacts present in highly-compressed MPEG-2 reconstructed videos. The artifacts mixtures, *C*, (blocky-blurry artifacts) were obtained using a fixed combination of a pure blurry and blocky artifacts given by the following equation:

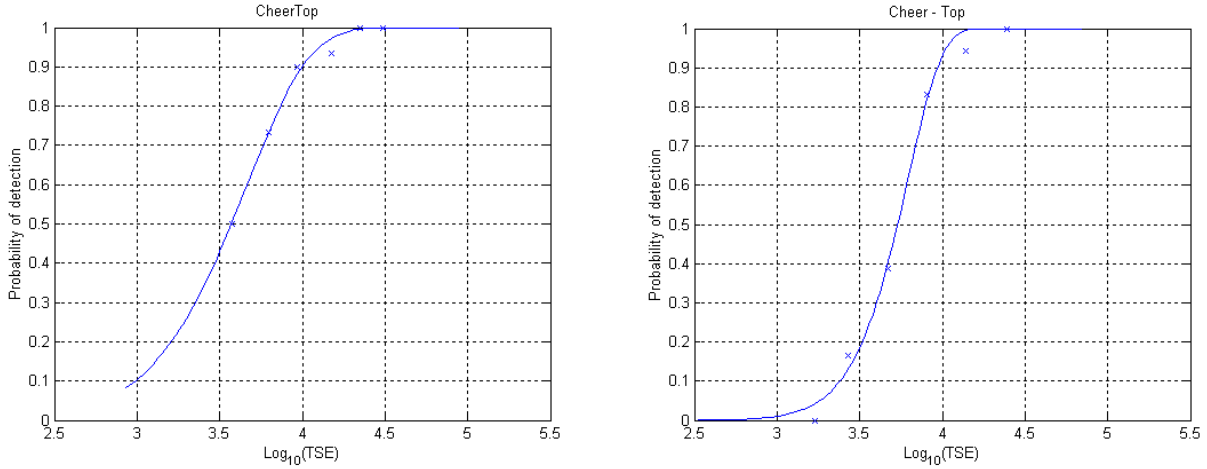
$$C = 0.5 \cdot A_{blocky} + 0.5 \cdot A_{blurry}, \quad (7)$$

where $\boldsymbol{A}_{blocky}$ is the sequence with block artifacts, $\boldsymbol{A}_{blurry}$ is the sequence with blurring artifacts. This combination rule was chosen based mostly on the basis of visual comparisons. We searched for a proportion that looked as similar as possible to a MPEG-2 sequence compressed at 7.5 Mbps, then reconstructed. We also measured the averages and variances of the luminance of the created video sequences to assure a rough match with the MPEG-2 reconstructed ones. This combination rule produced artifacts that were the most perceptually similar to MPEG-2 artifacts compressed to 7.5 Mbps. They also roughly matched the MPEG-2 artifacts in the mean and variance of their luminance distributions.

Data analysis

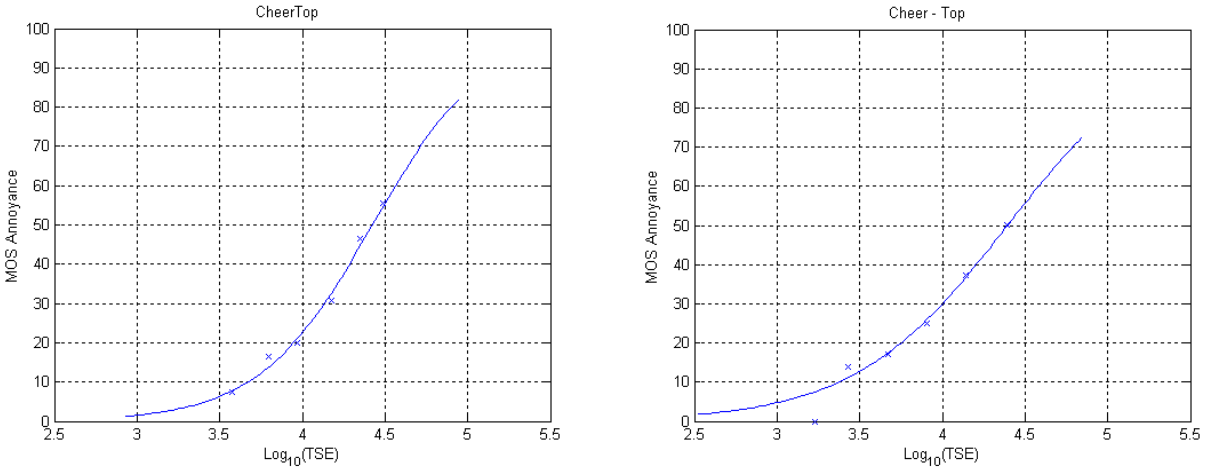
Figure 4 depicts the psychometric functions for both the synthetic artifacts and the MPEG-2 artifacts coded at 7.5 Mbps. The video is Cheerleaders with the defect zone filling the top one third of the frame. As expected, the curves have the same shape.

The parameters *S* and *k*, for all test sequences, are shown in Table 2 for both synthetic and MPEG artifacts. The empty spaces in Table 2 correspond to cases where a fit was not possible. The MPEG-2 data are from Reference [5]. The sensitivity parame -



(a) (b)

Figure 4. Probability of detection curves for (a) synthetic artifacts, (b) MPEG-2 artifacts, corresponding to the video Cheerleaders with the defect zone filling the top part of the frame.



(a) (b)

Figure 5. Mean annoyance curves for (a) synthetic artifacts, (b) MPEG-2 artifacts, corresponding to the video Cheerleaders with the defect zone filling the top part of the frame.

ters, *S*, for both types of artifacts are highly correlated. The small differences in the parameter values for *S* and *k* are not statistically significant (T-test, two-tailed, $p > 0.05$).

Figure 5 depicts the mean annoyance values versus the log total squared error for the synthetic artifacts and the MPEG-2 artifacts coded at 7.5 Mbps, corresponding to the video Cheerleaders with the defect zone filling the top part of the frame. Again, the curves have similar form. Most of the annoyance functions for the synthetic artifacts are shifted slightly to the right implying that the same error produces slightly less annoyance. Also, most of these functions are steeper for the synthetic artifacts, implying that annoyance grows faster with TSE for these artifacts.

The parameters of the annoyance functions ($\overline{x}$ and β) are shown in Table 2. Here the small differences in the parameters for the corresponding MPEG-2 and synthetic artifacts are statistically significant. The values, for the two types of artifacts are highly correlated.

In summary, the synthetic artifacts look very realistic and have objective error measures within the same range as the MPEG-2 artifacts. A comparison of the results of this experiment with the results from [5] showed that the form of the functions for MPEG-2 and synthetic artifacts is the same and *S* and $\overline{x}$ parameters are highly correlated across the two artifact types. Thresholds are essentially the same, but synthetic artifacts are slightly less annoying than MPEG-2 artifacts at the same TSE.

Future work

For our future work, we intend to perform the following tasks:

1. Study other types of artifacts:
   We are interested on the following artifacts: ringing, edge busyness, noise errors, jerkiness, ghosts, image persistence and false colors. To study these artifacts, we will do a series of psychophysical experiments. We will use real and synthetic artifacts.
2. Design algorithms of measuring error measurements for the most important artifacts.
   We intend to investigate what are the best techniques to extract features from an image. These features will then be translated into error component measures, as described earlier.
3. Combine the error measurement in an overall objective measure
   The final part of this work is to combine all the measures into an one quality score. This is no simple addition problem. The combination rule will have to take into account no only the importance of each artifact and their masking/ properties.

Part II: Non-Reference Objective Quality Measurement System based on Data Hiding

Full-reference and reduced-reference objective metrics cannot be used in most practical transmission applications (internet, broadcast, cable, etc.) where the original video is not available. The cost of sending side information is very high and an automated assessment method not requiring any reference is needed. The goal of this part of our work is to design a non-reference method for estimating impairments introduced during capture, transmission, storage, and display as well as by some transformation like compression or enhancement. The proposed algorithm makes use of data hiding techniques to transmit the necessary reference information embedded in the video signal. Our preliminary work has shown that the degradation caused by compression and transmission can be estimated by detecting the extent of the marker's degradation. The method can be used regardless of the video content or application. The degradation caused by the embedding procedure can be made small enough so that the human eye cannot perceive it.

In this part of work we intend to develop such a system. Psychophysical experiments will be used with the goal of testing the correlation between this objective quality metric and the quality experienced by human viewers. We will also use the experiments limit the guarantee that this scheme does not introduce any further impairments to the videos.

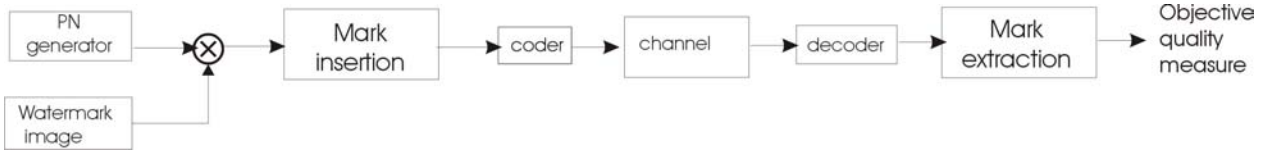


Figure 6 Basic scheme of tracing watermarking for coder-channel quality assessment in multimedia communications.

The embedding procedure

Several watermarking methods have been proposed in the literature; a first classification of these methods can be done according to the particular domain in which the embedding process is performed. The watermark insertion can be done in the spatial domain [17] or in an *ad hoc* transform domain such as the DCT domain [18], the Fourier domain [19], or the Wavelet domain [20], [21].

Figure 6 depicts the block diagram of the embedding system used in this work. A two-dimensional mark is embedded in the DCT domain. Figure 7 depicts the marks used so far in our test. A preprocessing is performed on the mark before the actual embedding into the DCT mid-frequency samples. A spread-spectrum technique is simulated to hide the mark, by using a set of uncorrelated pseudo-random noise (PN) matrices (one per each frame), which are later, multiplied by the reference watermark (the same for the whole video). The result is that the spatial localization of the mark is different frame by frame, avoiding temporal summation. For a generic frame of a video sequence $f_i(n_1, n_2)$, the final mark w_i is obtained by multiplying the mark $m(n_1, n_2)$ by $p_i(n_1, n_2)$:

$$w_i(n_1, n_2) = m(n_1, n_2) \cdot p_i(n_1, n_2) \tag{8}$$

The log of the luminance is taken and then the DCT coefficients are computed. The final mark is multiplied by a scaling factor α and is added to the DCT coefficients:

$$Y_i(n_1, n_2) = \mathrm{DCT}(\log(f_i(n_1, n_2))) + \alpha \cdot w_i(n_1, n_2) \tag{8}$$

The mark is added only to the mid-frequencies DCT coefficients. The range of frequencies where the watermark is inserted is strongly dependent on the application. For the purpose of delivering a high quality video through an ideal channel, the mid-frequencies are a good choice. Inserting the mark in the low-frequencies would cause visible artifacts in the image, while inserting it in the high frequencies would make it easier to remove it. This scaling factor α can be used to vary the strength of the watermark. In various applications and video formats different values for α are desirable. By increasing α, we also decrease the quality of the video.



(a) (b)

Figure 7. Mark images used in our embedding scheme: (a) Logo and (b) Random.

Work done so far

We have implemented the system described in the previous section. The current system embeds a regular gray-level image into the video. We intend to create new images to be embedded which are more adequate to the task of measuring quality. Other improvements in the algorithm itself, like changing the area in which the watermark is embedded, are still being investigated.

To be useful, the embedding system cannot introduce visible artifacts into the images. With the objective of measuring the detection threshold and annoyance values for the artifacts introduced by the watermark embedding procedure, we performed a psychophysical experiment using the techniques described in Section [22]. We used twenty test subjects for this experiment. To generate the test video sequences, we start with a set of five original video sequences of assumed high quality: "Bus", "Cheerleader", "Flower-garden", "Football" and "Hockey". The total number of test sequences used in this experiment is 65, which includes 60 test sequences (5 originals times 6 strength factors times 2 watermark images) plus the 5 original sequences.

To find artifact detection and annoyance values, the contrast of the error patterns must range from nearly imperceptible to highly annoying. This is obtained by varying the scaling factor used in the watermark embedding algorithm. The set of scaling factors used is 0.1, 0.2, 0.3, 0.4, 0.5, and 0.6. Figure 8 displays a detail of 10th frame of the video Cheerleaders with (a) and



(a) (b)

Figure 8. Detail of the cheerleader sequence (a) without and (b) with mark Logo with $\alpha = 0.6$.

without (b) the mark. The mark embedded in this case was "Logo" with a scaling factor $\alpha = 0.6$, which is considered very strong.

Figures 9 depicts the psychometric curve for the artifacts caused by embedding the watermark image "Logo" (a) and "Random" (b), respectively, into the video Cheerleaders. The two curves are very similar implying that the choice of a different image as a watermark does not have a significant effect on the visibility of the artifacts.

Table 3 summarizes the 50% detection threshold found in terms of error energy (TSE) and logarithmic error energy ($\log_{10}$(TSE)). Overall, the threshold values do not change considerably over the sequences and remain almost constant when only the embedded image is changed. Table 3 also includes the curve fit parameters found for each sequence and image tested. The parameter S (sensitivity) corresponds to the inverse of the log-threshold and therefore has the same behavior.

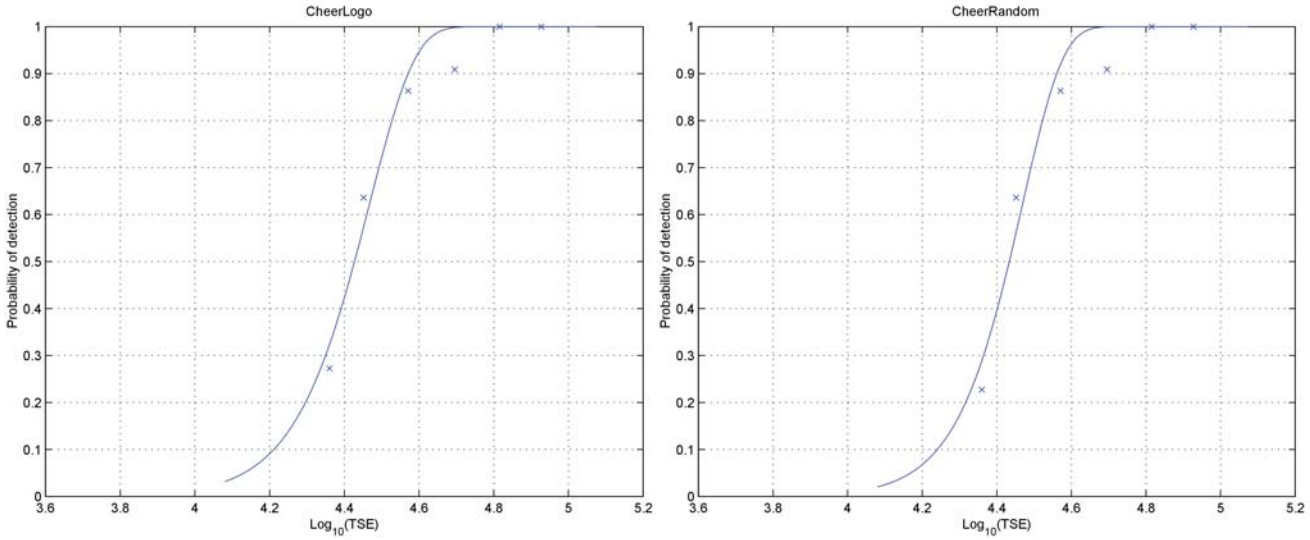
The parameter k, which represents the steepness of the probability of detection curve, varies between different videos and may be due to variation in content of the videos. The same artifact at the same strength will vary both in visibility and annoyance depending on where and when it appears, and also on the texture and luminance characteristics of the background of the video. In particular, it is noticed that white and/or smooth backgrounds facilitate the visibility of the artifacts. For the video Hockey, which has very large white smooth areas, it has not been possible to estimate the detection threshold. More than half of all subjects tested have seen the weakest artifact. This particular video has probably a lower threshold than the rest of the videos. It has not also been possible to calculate the threshold for the video Flower embedded with the watermark Logo due to artifacts (blurring) previously present in this video.

The results [22] show that the choice of the image to be embedded into the video does not affect the visibility and annoyance of the artifacts significantly. The mean annoyance curve can vary considerably depending on the physical characteristics of the particular video. The detection thresholds found are indicators of the strength of the watermark, which is seen by 50% of the observers. In our experiment, the thresholds found correspond to scaling factors α no greater than 0.2.

We are also interested in analyzing what other types of artifacts are introduced by the embedding. One interesting idea for this type of system would be to implement the embedding procedure at the same time as the compression is being made. This ap-

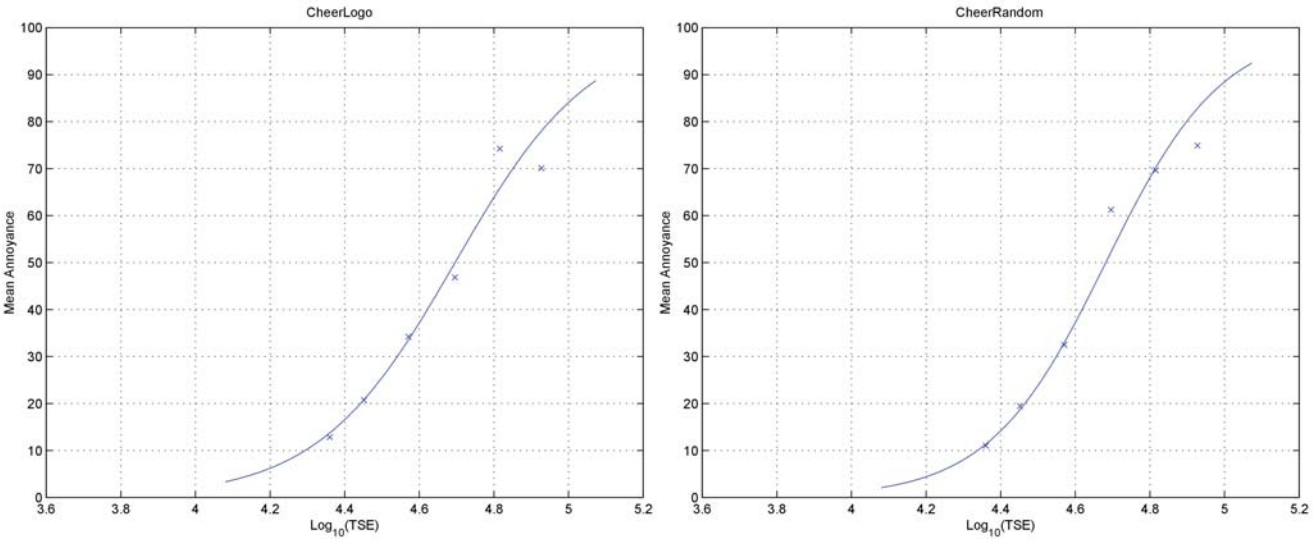
Table 3. Threshold error energy and detection threshold curve fit parameters for the artifacts.

	Detection Threshold		Curve Fit Parameters	
Test Sequences	**TSE**	**Log_{10}(TSE)**	**S**	**K**
Bus-Logo	15488	4.19	0.2407	27.71
Bus-Random	16218	4.21	0.2378	31.17
Cheer-Logo	27542	4.44	0.2259	37.66
Cheer-Random	28184	4.45	0.2256	42.38
Flower-Random	39811	4.60	0.2179	87.50
Foot-Logo	26303	4.42	0.2270	25.68
Foot-Random	26303	4.42	0.2273	27.38



(a) (b)

Figure 9. Psychometric curve for embedding the mark (a) Logo and (b) Random into the video cheerleaders.



(a) (b)

Figure 10. Mean annoyance curve for embedding the mark (a) Logo and (b) Random into video cheerleaders **.**

proach may introduce color artifacts into the video. Although the embedding scheme produces only imperceptible changes in the luminance component, color artifacts may be caused due to errors in the motion estimation and prediction stages. Experimental results show that these artifacts can become fairly noticeable as the strengths of the watermark are increases.

Future Work

As future work, we intend to perform the following tasks:

1. Improvement of data hiding system

Some improvements to the embedding procedure will be made. It is important to make sure that the algorithm is fast and reliable. An ideal situation would be to have the marks embedded during the compression/ coding.

2. Design a perceptual mark

We believe it is possible to design a better mark, making use of the characteristics of human vision system. This mark would the characteristic of being invisible and yet being able to any degradation the video might suffer.

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Timeline

Summer of 2002 (July-September)	Fall 2002 (October-December)	Winter 2003 (January-March)
Study ringing and edge busyness artifacts: - Perform experiments to test data - Analyze data Write the introduction and the outline of the thesis (maybe the beginning of some other chapter)	Study color artifacts: - Perform experiments to test data - Analyze data Design Psychophysical marks	Development of individual metrics for artifacts

Spring 2003 (April-June)	Summer 2003 (July-September)	Fall 2003 (October-December)
Development of overall full-reference metrics.	Write thesis	Thesis defense